

Welcome to ISE 201

Math Foundations for Decision & Data Science

- 1) Please note that we are recording **ALL** Class Zoom Sessions
- 2) Please complete skills assessment poll
- 3) **Session 01 Agenda:**
 - a) Introduction & Syllabus
 - b) Electronic Data Process & your project
 - c) Statistics Chapter 6 – Descriptive Statistics
 - d) Class project

Khaled Mabrouk (professional)

- Industrial Engineering Intern, **GM** Delco Remy Division, Anderson, IN
- Experienced Staff Consultant (Programmer), **Accenture**, Chicago
- Regional Account Manager, **Systems Modeling Corporation**, State College PA, Pittsburgh PA, and Ann Arbor, MI
- Simulation Engineer (entrepreneur), **The Model Builders**, Ann Arbor
- General Director Continuous Improvement, **Union Pacific Railroad**, Omaha, Nebraska
- Senior Operational Excellence Specialist, **PayPal**, San Jose
- Simulation Engineer & Coach (entrepreneur), **Sustainable Productivity Solutions**, Watsonville

Khaled Mabrouk (academic)

- Designed & delivered following courses:

ISE 167 – Introduction to Simulation (SJSU)

ISE 265 – Advanced Simulation Modeling (SJSU)

Manufacturing Systems Simulation (Wayne State University)

Systems Simulation (Eastern Michigan University)

- Delivered following courses: (SJSU)

ISE 131 – Statistics for Statistical Process Control

ISE 142 – Service Systems Engineering

ISE 155 – Supply Chain Engineering

ISE 195 – Senior Design Projects

Course Learning Outcomes

The following are the Course Learning Outcomes (CLO). The class work/assignments that most support that CLO are ***highlighted in italic bold***.

- CLO 1. To use multi-dimensional arrays to represent data of independent variables and responses, including to learn about descriptive statistics and graphical methods to summarize data using statistical software (Python). ***(Week One and Project)***
- CLO 2. Review linear algebra topics and to manipulate arrays and matrices to discover variables-responses relationships. ***(Weeks Two & Three, Homework 2, Quiz 1 and Project)***
- CLO 3. To learn, master and explain basic probability concepts and its use in statistical inference. ***(Weeks Four & Five, Homework 3, Quiz 2 and Project)***
- CLO 4. To learn, master and explain basic concepts of statistical inference. ***(Weeks Six & Seven, Homework 4, Quiz 3 and Project)***
- CLO 5. To formulate General Linear Models, including regression for quantitative variables and analysis of variance for discrete variables and know and explain usefulness and limitations. ***(Weeks Eight Nine & Ten, Homework 4, Quiz 3 and Project)***
- CLO 6. To use statistical software to analyze data, reach statistical valid conclusions and explain them to peers and management. ***(throughout class and Project)***
- CLO 7. To understand and explain the usefulness and limitations of basic optimization techniques for linear and nonlinear regression. ***(Weeks Eight Nine & Ten, Homework 5, Quiz 4 and Project)***

Semester Schedule

ISE 201 Semester Schedule - Fall 2025

Week	Date	Topic	Textbook	Chapter	Homework Due	Quiz Due	Project Deliverable
01	10/7	Desriptive Statistics & EDA	Montgomery	6			
02	10/14	Linear Algebra	Kneusel	5,6			
03	10/21	Linear Algebra	Kneusel	5,6	HW 01 (assigned 10/7)		Proposal
04	10/28	Probability	Montgomery	3,4,5	HW 02 (assigned 10/21)	Quiz 01	
05	11/4	Probability	Montgomery	7,8			
06	11/12	Statistics	Montgomery	9,10	Homework 03 (assigned 11/4)	Quiz 02	
07	11/18	Statistics	Montgomery	11,12,13			EDA
08	11/25	Statistics	Montgomery				
09	12/2	Optimization	Hillier	3	Homework 04 (assigned 11/25)	Quiz 03	
10	12/9	Optimization	Hillier	7,8			
11	12/16	Review			Homework 05 (assigne 12/9)	Quiz 04	Podcast

Wednesday

Course Grading

- | | |
|--|------------|
| ➤ 5 Homework Assignments | 20% |
| ➤ 4 Quizzes (3 take home, 4th in-class) | 40% |
| ➤ Project Proposal | 10% |
| ➤ Project EDA | 10% |
| ➤ Project Podcast (presentation) | 20% |

Motivation for Decision Science Analysis

*Statistical data analysis can provide insight
that allows us to become
more successful decision makers,
problem solvers,
and process/product designers.*

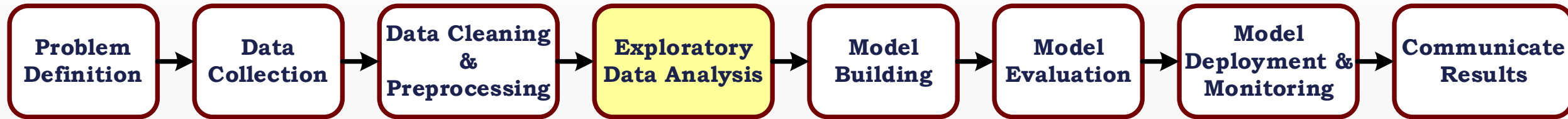
ISE 201 Purpose – Develop Your Skills to Utilize:

- Descriptive statistics and graphical methods to summarize and visualize data (understand data)
- Linear algebra concepts to understand variable-response relationships (matrices)
- Probability concepts to quantify risk in statistical inference (confidence intervals)
- Statistical inference to make decisions based on samples (hypothesis testing)
- Predictive models and optimization methods for finding the best approximation to the data (linear regression & simplex method)

Exploratory Data Analysis (EDA)

Your Data Science Project

EDA and Data Science Process



In Python:

- Pandas is used to clean, filter, and manipulate the data.
- Matplotlib is used to create basic visualizations.
- Seaborn is used to make better visualizations.

EDA Basics

- Goal of EDA is to clean and transform data for optimal learning by machine learning algorithms
- EDA is a process for exploring data sets through statistical analysis and graphics techniques to gain a deeper understanding of data.
- Objectives for EDA are to:
 - 1) Describe your data (variables)
 - 2) Closely explore data distributions (range & form of variability/patterns)
 - 3) Understand relationship between variables (types of correlation)
 - 4) Form data into groups (for further analysis)
 - 5) Notice unexpected patterns within & across groups (identify outliers & distinct group identifiers)
 - 6) Estimate Correlations and associations across groups (quantify levels of correlation)
 - 7) Visualize distributions, and relationships between variables & groups (arrive at insights)
 - 8) Choose best modeling techniques

EDA Process



EDA Step One – Understand Problem & Data

- What is business goal or research question we're interested in.
- There are plenty of open datasets available for us to use for ISE 201 project:

<https://datasetsearch.research.google.com>

<https://www.data.gov/>

<https://www.dataquest.io/blog/free-datasets-for-projects/>

- Review Bee Pollen Excel File

Data from: Identity and diversity of pollens collected by two managed bee species while in blueberry fields for pollination

Department of Agriculture — Pollen data from the manuscript titled: Identity and diversity of pollens collected by two managed bee species while in blueberry fields for pollination, published in...

XLS

Federal

Bee Pollen Excel File

Data.Gov > search “Entomology” > choose an excel file

DATA CATALOG

[Home](#) / [Datasets](#) [Organizations](#)

[Home](#) / [Department of Agriculture](#) / [Agricultural...](#)

[Feedback](#)



USDA

Department of Agriculture

There is no description for this organization

Publisher

Agricultural Research Service

Contact

Graham, Kelsey

Data from: Identity and diversity of pollens collected by two managed bee species while in blueberry fields for pollination

Metadata Updated: April 21, 2025

Pollen data from the manuscript titled: Identity and diversity of pollens collected by two managed bee species while in blueberry fields for pollination, published in Environmental Entomology. Pollen was collected from honey bee (*Apis mellifera*) and bumble bee (*Bombus impatiens*) forager returning to the colony, while in commercial highbush blueberry fields in SW Michigan for crop pollination. Pollen was identified visually using morphological features. Funding also provided by Project GREEN award GR18-038. Project GREEN, Michigan's plant agriculture initiative housed at Michigan State University, is a cooperative effort by plant-based commodity groups and businesses in cooperation with Michigan State University AgBioResearch, Michigan State University Extension and the Michigan Department of Agriculture and Rural Development to advance Michigan's economy through plant-based agriculture.

Access & Use Information

Public: This dataset is intended for public access and use.

License: us-pd

Bee Pollen Excel File

Bee species	Pollen collected from either honey bees (Apis mellifera) or bumble bees (Bombus impatiens), as indicated										
Year	Year of data collection										
Site code	Farm code name										
Date	Date sample was collected in the field										
hive_code	Unique hive code										
pollen_type	Pollen type as identified through morphology										
Pollen grains counted on slide	For honey bee samples, we counted pollen grains of each pollen type at 3 random locations on the slide, at 400x. All grains within the field of vision were counted.										
Total pollen grains counted on slide	For honey bees, the total number of pollen grains counted on that entire slide, in order to calculate the percent of each pollen type per slide.										
PercentSlide	For honey bees, calculated by dividing the pollen grains counted on slide for that pollen type by the total pollen grains counted on that slide, times 100. For bumble bees, the percent of each pollen type per slide was estimated.										

Bee Pollen Excel File

(let's look at Excel File)

	A	B	C	D	E	F	G	H	I	J
1	Bee species	Year	Site code	Date	hive_code	pollen_type	Pollen grains counted on slide	Total pollen grains counted on slide	PercentSlide	
560	Honey bees	2019	FarmA	6/11/2019	16.7	Prunus	63	300	21.00	
561	Honey bees	2019	FarmA	6/11/2019	16.7	Rhamnus_Frangula	98	300	32.67	
562	Honey bees	2019	FarmA	6/11/2019	16.7	Salix	72	300	24.00	
563	Honey bees	2019	FarmA	6/11/2019	16.4	Ornithogalum_umbel	24	300	8.00	
564	Honey bees	2019	FarmA	6/11/2019	16.4	Prunus	80	300	26.67	
565	Honey bees	2019	FarmA	6/11/2019	16.4	Rhamnus_Frangula	67	300	22.33	
566	Honey bees	2019	FarmA	6/11/2019	16.4	Salix	81	300	27.00	
567	Honey bees	2019	FarmA	6/11/2019	16.4	Trifolium_repens	48	300	16.00	
568	Honey bees	2019	FarmB	5/27/2019	4.4	Lindera_benzoin	23	300	7.67	
569	Honey bees	2019	FarmB	5/27/2019	4.4	Prunus	69	300	23.00	
570	Honey bees	2019	FarmB	5/27/2019	4.4	Rhamnus_Frangula	54	300	18.00	
571	Honey bees	2019	FarmB	5/27/2019	4.4	Salix	154	300	51.33	
572	Honey bees	2019	FarmB	5/27/2019	4.8	Lindera_benzoin	5	300	1.67	
573	Honey bees	2019	FarmB	5/27/2019	4.8	Prunus	77	300	25.67	
574	Honey bees	2019	FarmB	5/27/2019	4.8	Salix	218	300	72.67	
575	Honey bees	2019	FarmB	5/27/2019	4.2	Prunus	99	300	33.00	
576	Honey bees	2019	FarmB	5/27/2019	4.2	Rhamnus_Frangula	67	300	22.33	
577	Honey bees	2019	FarmB	5/27/2019	4.2	Salix	134	300	44.67	
578	Honey bees	2019	FarmB	5/30/2019	4.8	Lindera_benzoin	7	300	2.33	
579	Honey bees	2019	FarmB	5/30/2019	4.8	Prunus	46	300	15.33	
580	Honey bees	2019	FarmB	5/30/2019	4.8	Salix	247	300	82.33	
581	Honey bees	2019	FarmB	5/30/2019	4.2	Prunus	18	300	6.00	
582	Honey bees	2019	FarmB	5/30/2019	4.2	Ranunculus_hispidis	1	300	0.33	
583	Honey bees	2019	FarmB	5/30/2019	4.2	Salix	278	300	92.67	
584	Honey bees	2019	FarmB	5/30/2019	4.2	Trifolium_repens	3	300	1.00	
585	Honey bees	2019	FarmB	5/30/2019	4.4	Prunus	83	300	27.67	
586	Honey bees	2019	FarmB	5/30/2019	4.4	Ranunculus_hispidis	4	300	1.33	

EDA Step Two – Import & Inspect Data

- Load data into environment, while avoiding errors or truncations
- Check size of data, number of rows and columns, to understand complexity
- Check for missing values
- Identify data type for each variable, numerical, categorical, etc.
- Look for errors or inconsistencies such as invalid values, mismatched units, or outliers.

EDA Step Two – Import & Inspect Data

```
: import pandas as pd
import numpy as np

BPdata = pd.read_excel('BeesPollen.xlsx', engine='openpyxl')

print(BPdata.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5530 entries, 0 to 5529
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Bee species                           5530 non-null   object
1   Year                                  5530 non-null   int64
2   Site code                             5530 non-null   object
3   Date                                  5530 non-null   datetime64[ns]
4   hive_code                             5530 non-null   float64
5   pollen_type                           5530 non-null   object
6   Pollen grains counted on slide         781 non-null    float64
7   Total pollen grains counted on slide    781 non-null    float64
8   PercentSlide                           5530 non-null   float64
dtypes: datetime64[ns](1), float64(4), int64(1), object(3)
memory usage: 389.0+ KB
None
```

```
import pandas as pd
import numpy as np

BPdata = pd.read_excel('BeesPollen.xlsx', engine='openpyxl')

print(BPdata.head())
```

	Bee species	Year	Site code	Date	hive_code	pollen_type \
0	Honey bees	2018	FarmB	2018-05-29	1.0	Salix
1	Honey bees	2018	FarmB	2018-05-29	1.0	Rhamnus_Frangula
2	Honey bees	2018	FarmB	2018-05-29	1.0	Lonicera
3	Honey bees	2018	FarmB	2018-05-29	1.0	Prunus
4	Honey bees	2018	FarmB	2018-06-01	1.0	Rhamnus_Frangula

	Pollen grains counted on slide	Total pollen grains counted on slide \
0	357.0	388.0
1	29.0	388.0
2	1.0	388.0
3	1.0	388.0
4	71.0	114.0

	PercentSlide
0	92.010309
1	7.474227
2	0.257732
3	0.257732
4	62.280702

EDA Step Two – Import & Inspect Data

```
import pandas as pd
import numpy as np

BPdata = pd.read_excel('BeesPollen.xlsx', engine='openpyxl')
print(BPdata.shape)
```

(5530, 9)

```
import pandas as pd
import numpy as np

BPdata = pd.read_excel('BeesPollen.xlsx', engine='openpyxl')

print("Mean =", BPdata['PercentSlide'].mean())
print("Standard Deviation =", BPdata['PercentSlide'].std())
print("Median =", BPdata['PercentSlide'].median())
print("Mode =", BPdata['PercentSlide'].mode())
print("Minimum =", BPdata['PercentSlide'].min())
print("Maximum =", BPdata['PercentSlide'].max())
print("Count =", BPdata['PercentSlide'].count())
print("Length =", len(BPdata['PercentSlide']))
```

```
Mean = 47.60757685353508
Standard Deviation = 37.75596774109108
Median = 40.0
Mode = 0    100.0
Name: PercentSlide, dtype: float64
Minimum = 0.1
Maximum = 100.0
Count = 5530
Length = 5530
```

EDA Step Three – Handle Missing Data

- Understand patterns and possible causes
- Decide whether to remove missing data or fill in missing values
- Use appropriate imputation methods based on the data's characteristics
- Be attuned to impact of missing data

```
import pandas as pd
import numpy as np

BPdata = pd.read_excel('BeesPollenSHORT.xlsx')
df = pd.DataFrame(BPdata)

missing_values = df.isnull()
print(missing_values)
```

NOTE: PANDAS excludes missing data (by default) for all Descriptive Statistics

	Bee species	Year	Site code	Date	hive_code	pollen_type \
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
..	...	...	...	...	...	...
95	False	False	False	False	False	False
96	False	False	False	False	False	False
97	False	False	False	False	False	False
98	False	False	False	False	False	False
99	False	False	False	False	False	False

	Pollen grains counted on slide	Total pollen grains counted on slide \
0	False	False
1	False	False
2	False	False
3	False	False
4	False	False
..	...	...
95	False	False
96	False	False
97	False	False
98	False	False
99	False	False

	PercentSlide
0	False
1	False
2	False
3	False
4	False
..	...
95	False
96	False
97	False
98	False
99	False

[100 rows x 9 columns]

EDA Step Three – Handle Missing Data

Some Python Tools

“NA” is terminology for Missing Data for a Field (column) a specific Record (row)

- `datafilename.dropna` – removes records with no information
- `datafilename.fillna(replacement value)` – replace empty field with the replacement value
- `datafilename.drop_duplicates` – removes duplicates in file

EDA Step Four – Explore Data Patterns

```
import pandas as pd
import numpy as np

BPdata = pd.read_excel('BeesPollen.xlsx', engine='openpyxl')
print(BPdata.describe())
```

	Year	Date	hive_code \
count	5530.000000	5530	5530.000000
mean	2018.782821	2019-03-09 18:57:25.063291136	3.149765
min	2018.000000	2018-05-17 00:00:00	1.000000
25%	2019.000000	2019-05-14 00:00:00	2.000000
50%	2019.000000	2019-05-27 00:00:00	3.000000
75%	2019.000000	2019-06-03 00:00:00	4.000000
max	2019.000000	2019-06-12 00:00:00	22.400000
std	0.412363	NaN	3.003949

	Pollen grains counted on slide	Total pollen grains counted on slide \
count	781.000000	781.000000
mean	52.878361	236.402049
min	1.000000	42.000000
25%	3.000000	161.000000
50%	21.000000	300.000000
75%	80.000000	300.000000
max	431.000000	468.000000
std	71.655653	90.174999

	PercentSlide
count	5530.000000
mean	47.607577
min	0.100000
25%	10.000000
50%	40.000000
75%	90.000000
max	100.000000
std	37.755968

EDA Step Five – Transform Data

(Matrix Discussion in Linear Algebra Sessions)

- Scale/Normalize numerical variables
- Encode categorical variables for machine learning
- Apply mathematical transformations
- Create new variables from existing ones; i.e. calculate ratios or combine variables
- Aggregate/group data based on specific variables or conditions

EDA Step Six – Visualize Correlations

- ➊ For Categorical variables, create frequency tables, bar plots and pie charts
- ➋ For numerical variables, generate histograms, box plots, violin plots, and density plots
- ➌ To find relationships between variables, use scatter plots, correlation matrices, and statistical tests

EDA Step Six – Visualize Correlations

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
import seaborn as sns
import matplotlib.pyplot as plt

file_path = "BeesPollenSKNY.xlsx"

BPdata = pd.read_excel(file_path, engine='openpyxl')
df = pd.DataFrame(BPdata)

mapping = {'FarmA': 1, 'FarmB': 2, 'FarmC': 3, 'FarmD': 4, 'FarmE': 5, 'FarmF': 6, 'FarmG': 7, 'FarmHH': 8, 'FarmI': 9,
           'FarmJ': 10, 'FarmK': 11, 'FarmL': 12, 'FarmM': 13, 'FarmN': 14, 'FarmO': 15, 'Farm1': 16, 'Farm2': 17, 'Farm3': 18,
           'Farm4': 19, 'Farm5': 20, 'Farm6': 21, 'Farm7': 22}
df['Site code'] = df['Site code'].map(mapping)

correlation_matrix = df.corr()

print("\nCorrelation Matrix:")
print(correlation_matrix)

plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Matrix Heatmap")
plt.show()
```

EDA Step Six – Visualize Correlations

(Is value greater than 0.7?)

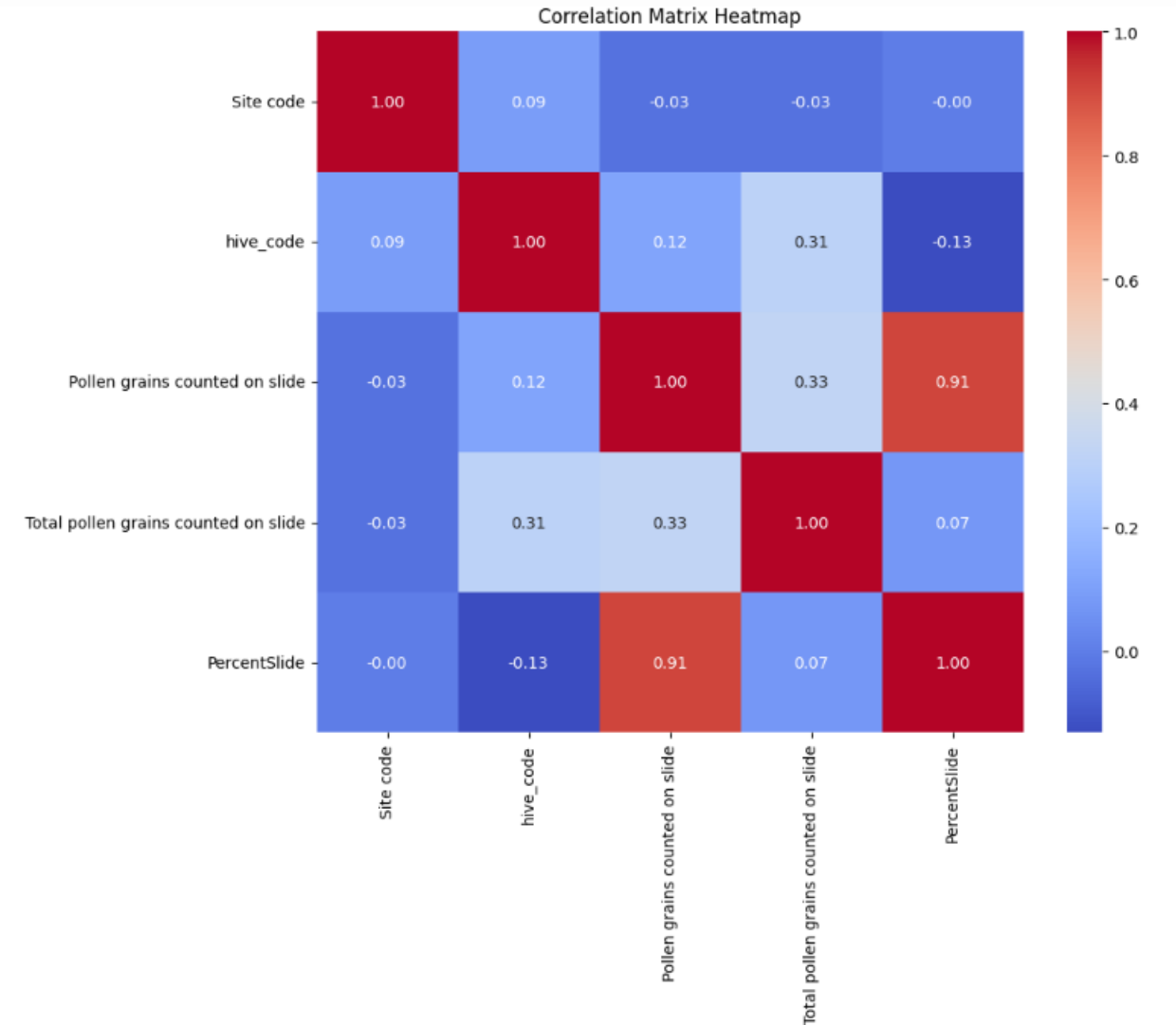
Correlation Matrix:

	Site code	hive_code	\
Site code	1.000000	0.089693	
hive_code	0.089693	1.000000	
Pollen grains counted on slide	-0.029836	0.115417	
Total pollen grains counted on slide	-0.032750	0.307024	
PercentSlide	-0.002454	-0.131489	

	Pollen grains counted on slide	\
Site code	-0.029836	
hive_code	0.115417	
Pollen grains counted on slide	1.000000	
Total pollen grains counted on slide	0.325734	
PercentSlide	0.912754	

	Total pollen grains counted on slide	\
Site code	-0.032750	
hive_code	0.307024	
Pollen grains counted on slide	0.325734	
Total pollen grains counted on slide	1.000000	
PercentSlide	0.074262	

	PercentSlide	\
Site code	-0.002454	
hive_code	-0.131489	
Pollen grains counted on slide	0.912754	
Total pollen grains counted on slide	0.074262	
PercentSlide	1.000000	



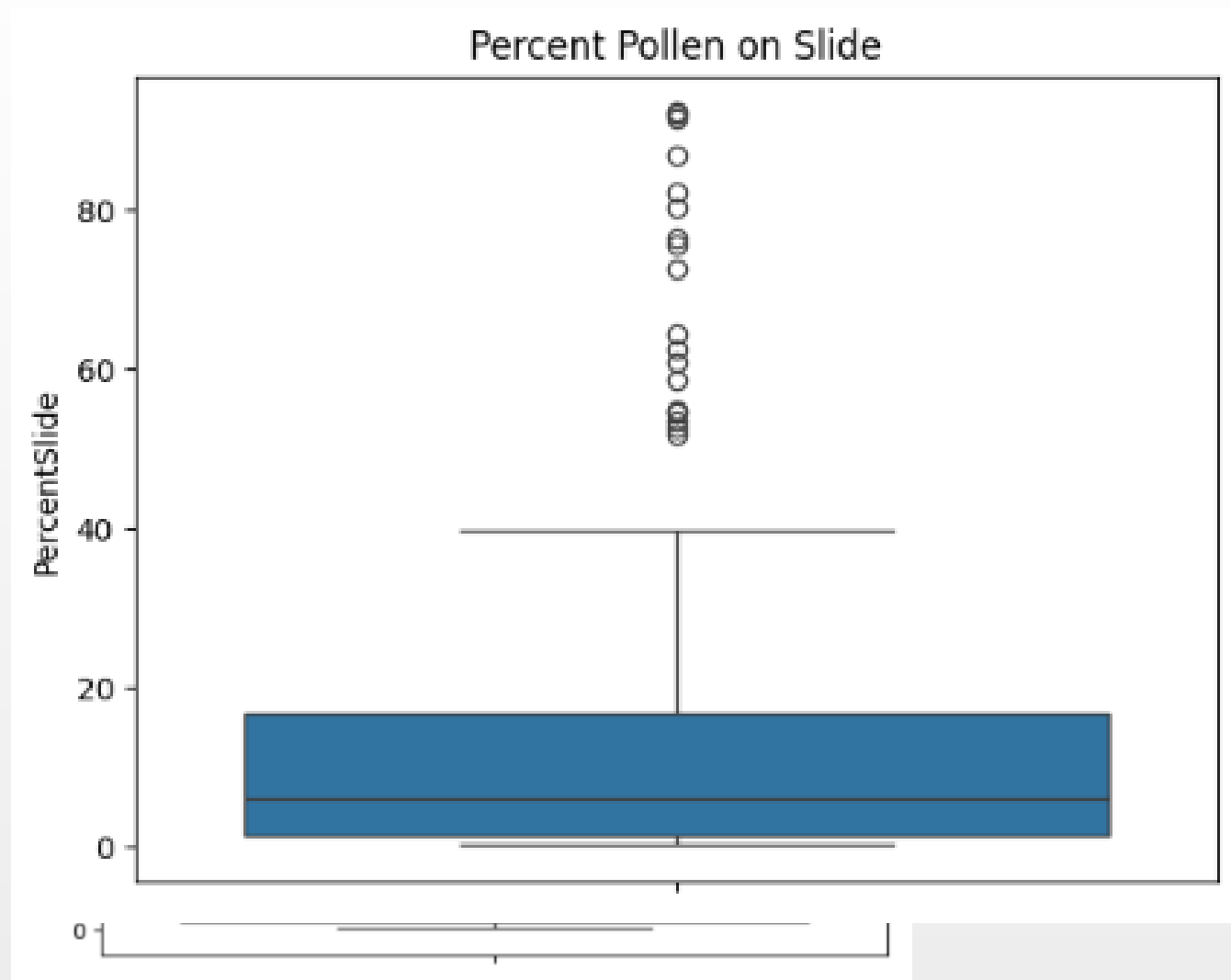
EDA Step Seven – Handle Outliers

- ➊ Identify Outliers
- ➋ Determine whether it should be removed or adjusted

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

BPdata = pd.read_excel('BeesPollenSHORT.xlsx')

sns.boxplot(data=BPdata['PercentSlide'])
plt.title('Percent Pollen on Slide')
plt.show()
```



EDA Step Eight – Communicate Insights

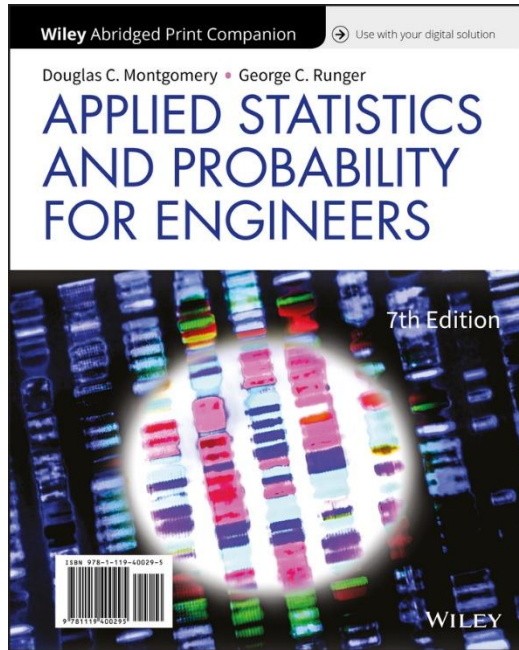
(& some insight into your project's final deliverable)

- Clearly state the goals and scope of your analysis
- Review steps taken to prepare data for analysis & perform analysis
- Use visualizations to support findings
- Highlight key insights, patterns, or anomalies
- Mention any limitations to the analysis
- Suggest next steps in investigation

Breakrooms Exercise

Who's using data in your
organization & How are
they using it

WILEY



Applied Statistics and Probability for Engineers

Seventh Edition

Douglas C. Montgomery George C. Runger

Chapter 6 Descriptive Statistics

Learning Objectives for Chapter 6

After **careful study** of this chapter, you should be able to do the following:

1. Compute and interpret the sample mean, variance, standard deviation, median, and range
2. Explain the concepts of sample mean, variance, population mean, and population variance
3. Construct and interpret visual data displays, including stem-and-leaf display, histogram, and the box plot
4. Explain the concept of random sampling
5. Construct and interpret normal probability plots
6. Explain how to use box plots and other data displays to visually compare two or more samples of data
7. Know how to use simple time series plots to visually display the important features of time-oriented data
8. Know how to construct and interpret scatter diagrams of two or more variables

Numerical Summaries of Data

- Data summaries and displays are essential to good statistical thinking
- It is useful to describe data features **numerically**
- Characterizing the **location** or **central tendency** in the data is an example of a numerical summary
- Data are often a **sample** of observations that have been selected from some larger **population** of observations
 - This type of population is called a **conceptual** or **hypothetical** population because it does not physically exist

**A sample has to
be representative
& random**

Sample Mean

The location or central tendency in the data can be characterized by the **arithmetic average** or the **sample mean**.

Sample Mean

If the n observations in a sample are denoted by $x_1, x_2, \dots, x_n$, the **sample mean** is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n} \quad (6.1)$$

For a finite **population** with N equally likely values, the probability mass function is $f(x_i) = 1/N$ and the mean is

$$\mu = \sum_{i=1}^N x_i f(x_i) = \frac{\sum_{i=1}^N x_i}{N}$$

Example 6.1 | Sample Mean

Consider 8 observations (x_i) of pull-off force from engine connectors as shown in the table.

$$\begin{aligned}\bar{x} = \text{average} &= \frac{\sum_{i=1}^8 x_i}{8} = \frac{12.6 + 12.9 + \dots + 13.1}{8} \\ &= \frac{104}{8} = 13.0 \text{ pounds}\end{aligned}$$

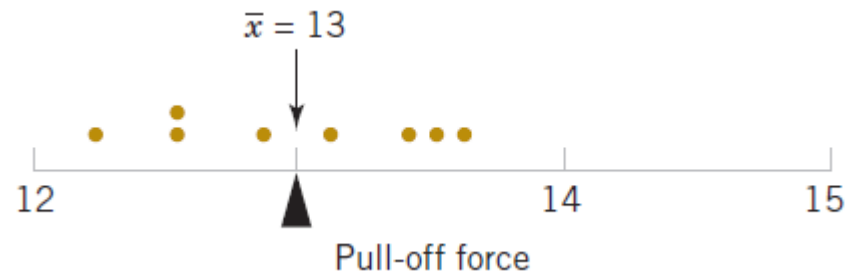


Figure 6.1 The sample mean is the balance point.

i	x_i
1	12.6
2	12.9
3	13.4
4	12.3
5	13.6
6	13.5
7	12.6
8	13.1
	13.00
= AVERAGE(\$B2:\$B9)	

Sample Variance and Standard Deviation

The variability or scatter in the data may be described by the **sample variance** or the **sample standard deviation**.

The units of measurement for the sample variance are the square of the original units of the variable, while the standard deviation measures variability in the original units.

Sample Variance and Standard Deviation

If $x_1, x_2, \dots, x_n$ is a sample of n observations, the **sample variance** is

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \quad (6.3)$$

The **sample standard deviation**, s , is the positive square root of the sample variance.

Example 6.2 | Sample Variance

The table displays the quantities needed for calculating the sample variance and sample standard deviation.

The numerator of s^2 is

$$\sum_{i=1}^8 (x_i - \bar{x})^2 = 1.60$$

so the sample variance is

$$s^2 = \frac{1.60}{8-1} = \frac{1.60}{7} = 0.2286 \text{ (pounds)}^2$$

and the sample standard deviation is

$$s = \sqrt{0.2286} = 0.48 \text{ pounds}$$

TABLE 6.1		Calculation of Terms for the Sample Variance and Sample Standard Deviation	
i	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	12.6	-0.4	0.16
2	12.9	-0.1	0.01
3	13.4	0.4	0.16
4	12.3	-0.7	0.49
5	13.6	0.6	0.36
6	13.5	0.5	0.25
7	12.6	-0.4	0.16
8	13.1	0.1	0.01
Total	104.0	0.0	1.60

The meaning of $n - 1$ in the denominator

- The population variance is calculated with N , the population size. Why isn't the sample variance calculated with n , the sample size?
- The true variance is based on data deviations from the true mean, μ .
- The sample calculation is based on the data deviations from $\bar{x}$, not μ .
- $\bar{x}$ is an estimator of μ ; close but not the same.
- So the $n - 1$ divisor is used to compensate for the error in the mean estimation.

Degrees of Freedom

- When the sample variance is calculated with the quantity $n - 1$ in the denominator, the quantity $n - 1$ is called the **degrees of freedom**
- Origin of term:
 - There are n deviations from the $\bar{x}$ in the sample
 - The sum of the deviations is zero
 - $n - 1$ of the observations can be freely determined but the n^{th} observation is fixed to maintain the zero sum

Sample Range

In addition to the sample variance and sample standard deviation, the sample range is a useful measure of variability.

Sample Range

If the n observations in a sample are denoted by $x_1, x_2, \dots, x_n$, the **sample range** is

$$r = \max(x_i) - \min(x_i) \quad (6.6)$$

For Example 6.3 (pull-off force data), the sample range is $r = 13.6 - 12.3 = 1.3$.

Using Python for Mean, Standard Deviation, etc.

```
import pandas as pd
import numpy as np

BPdata = pd.read_excel('BeesPollen.xlsx', engine='openpyxl')

print("Mean =",BPdata['PercentSlide'].mean())
print("Standard Deviation =",BPdata['PercentSlide'].std())
print("Median =",BPdata['PercentSlide'].median())
print("Mode =",BPdata['PercentSlide'].mode())
print("Minimum =",BPdata['PercentSlide'].min())
print("Maximum =",BPdata['PercentSlide'].max())
print("Count =",BPdata['PercentSlide'].count())
print("Length =",len(BPdata['PercentSlide']))
```

```
Mean = 47.60757685353508
Standard Deviation = 37.75596774109108
Median = 40.0
Mode = 0    100.0
Name: PercentSlide, dtype: float64
Minimum = 0.1
Maximum = 100.0
Count = 5530
Length = 5530
```

Breakrooms Exercise

**Generate Descriptive
Statistics (using Python)
For Fruit Prices Data File**

Frequency Distributions and Histograms

- A **frequency distribution** is a more compact summary of data than a stem – and – leaf diagram
- To construct, we must divide the range of the data into intervals, which are usually called **class intervals**, **cells**, or **bins**
- Choosing **number of bins** approximately equal to the square root of the number of observations often works well in practice

Frequency Distribution Table

TABLE 6.4 Frequency Distribution for the Compressive Strength Data in Table 6.2

Class	$70 \leq x < 90$	$90 \leq x < 110$	$110 \leq x < 130$	$130 \leq x < 150$	$150 \leq x < 170$
Frequency	2	3	6	14	22
Relative frequency	0.0250	0.0375	0.0750	0.1750	0.2750
Cumulative relative frequency	0.0250	0.0625	0.1375	0.3125	0.5875
	$170 \leq x < 190$	$190 \leq x < 210$	$210 \leq x < 230$	$230 \leq x < 250$	
Frequency	17	10	4	2	
Relative frequency	0.2125	0.1250	0.0500	0.0250	
Cumulative relative frequency	0.8000	0.9250	0.9750	1.0000	

Histograms

- A **histogram** is a visual display of the frequency distribution
- Provides a visual impression of the shape and distribution of the measurements and information about the central tendency and scatter or dispersion in the data
- **Unequal bin widths** will be employed

$$\text{Rectangle height} = \frac{\text{bin frequency}}{\text{bin width}}$$

Constructing a Histogram (Equal Bin Widths)

- (1) Label the bin (class interval) boundaries on a horizontal scale.
- (2) Mark and label the vertical scale with the frequencies or the relative frequencies.
- (3) Above each bin, draw a rectangle where height is equal to the frequency (or relative frequency) corresponding to that bin.

Histograms

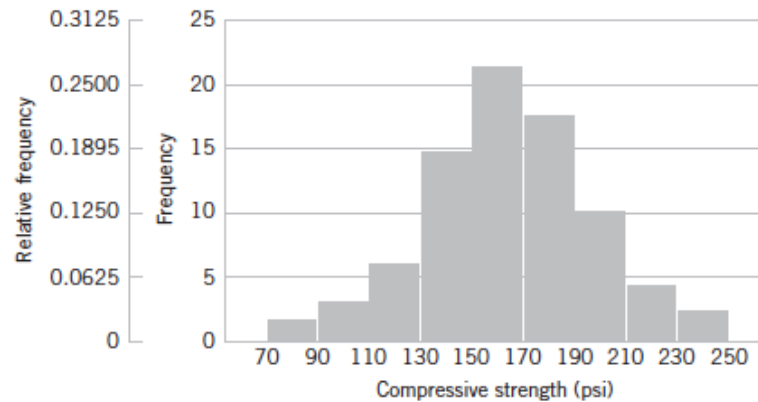


FIGURE 6.7
Histogram of compressive strength for 80 aluminum-lithium alloy specimens.

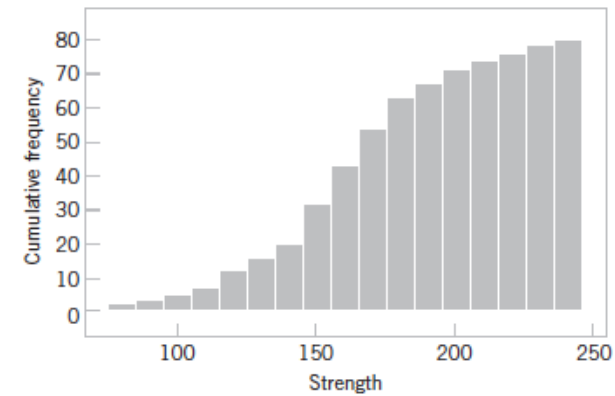


FIGURE 6.10
A cumulative distribution plot of the compressive strength data.

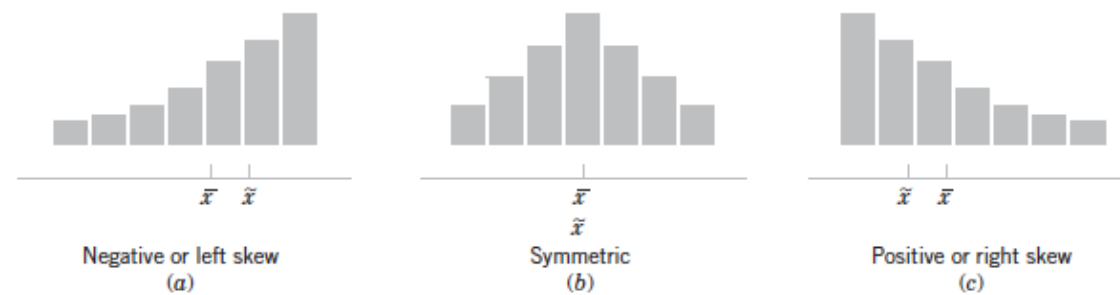


FIGURE 6.11
Histograms for symmetric and skewed distributions.

WILEY

Using Python to generate histogram

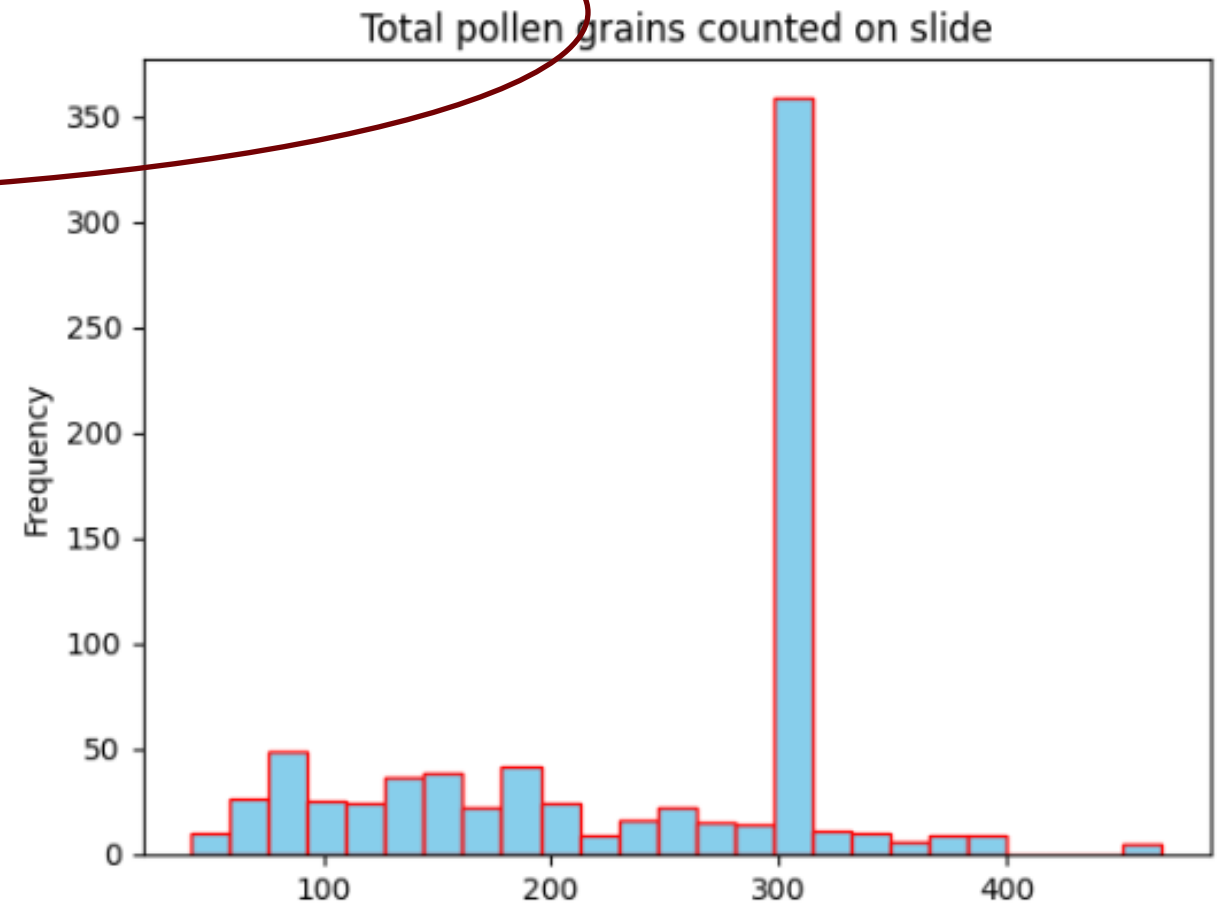
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

BPdata = pd.read_excel('BeesPollen.xlsx')

plt.hist(BPdata['Total pollen grains counted on slide'],bins=25,color='skyblue',edgecolor='red')

plt.xlabel('Values')
plt.ylabel('Frequency')
plt.title('Total pollen grains counted on slide')

plt.show()
```



Box Plots

- The **box plot** is a graphical display that simultaneously describes several important features of a data set, such as center, spread, departure from symmetry, and identification of unusual observations or outliers
- Sometimes called *box – and – whisker* plots
- Displays three quartiles
- A line, or **whisker**, extends from each end of the box

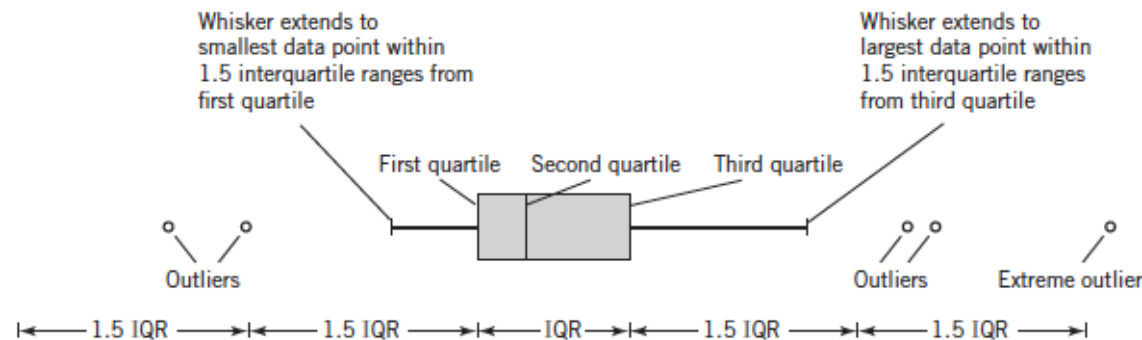


FIGURE 6.13

Description of a box plot.

Box Plots

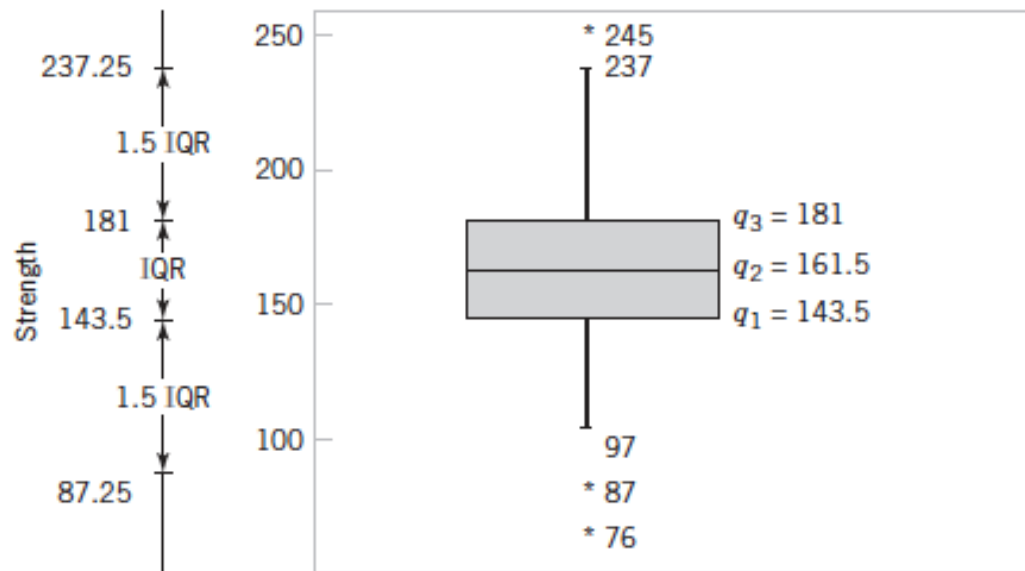


FIGURE 6.14

Box plot for compressive strength data in Table 6.2.

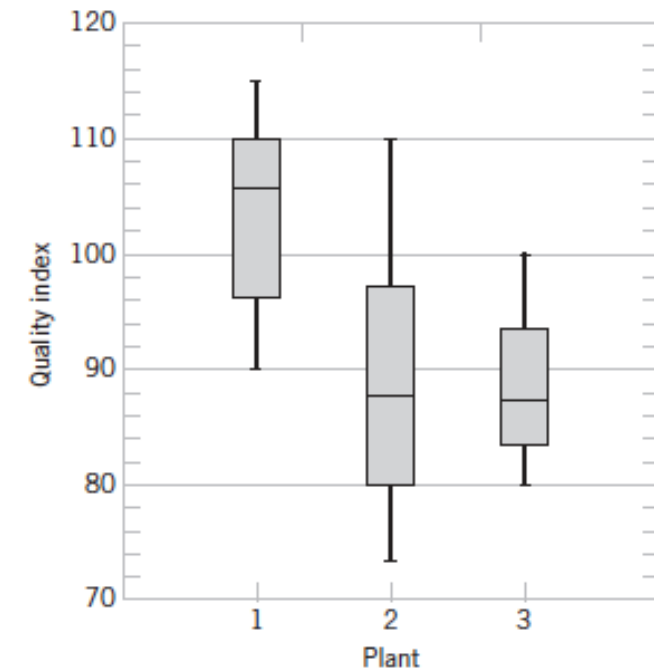


FIGURE 6.15

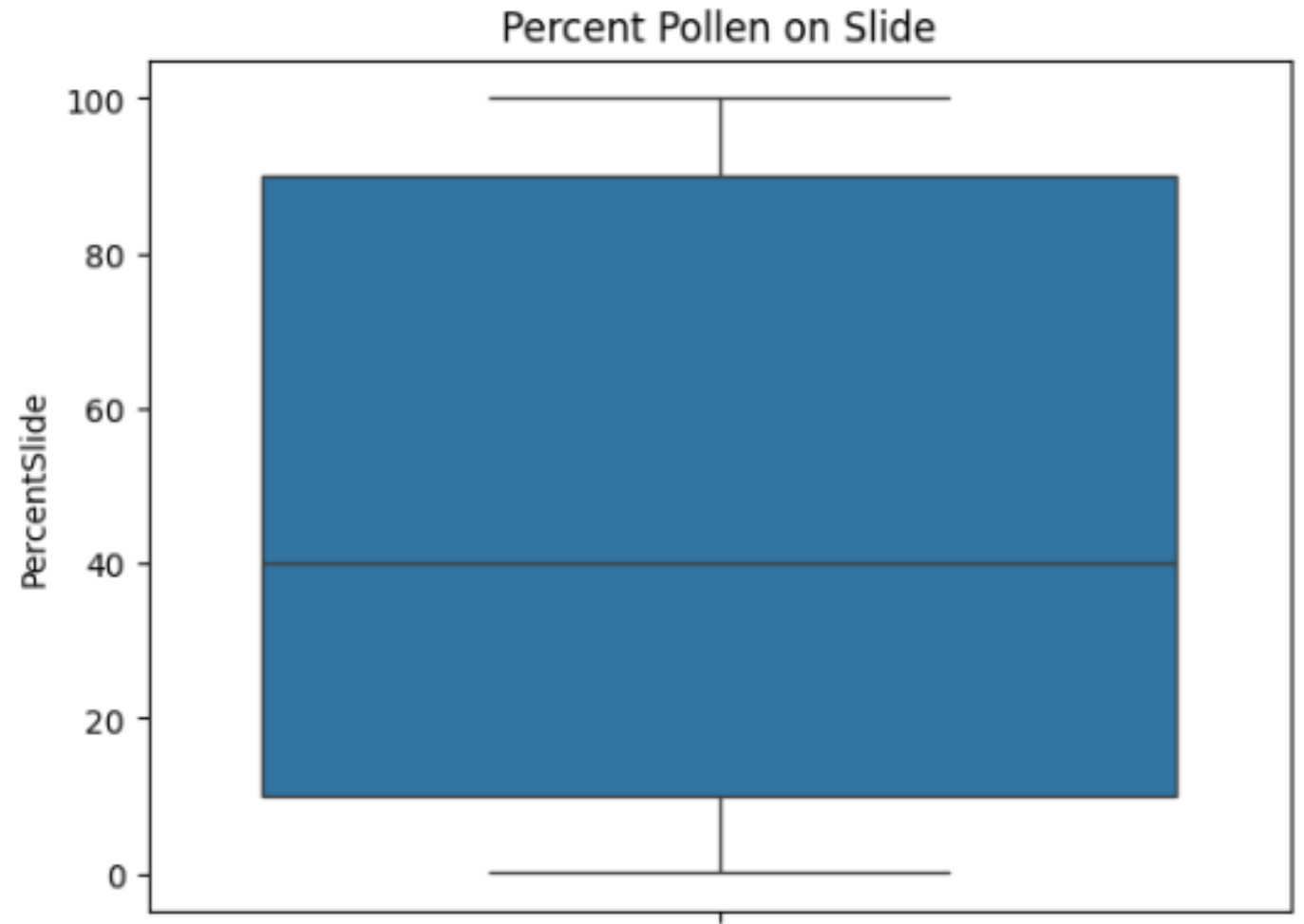
Comparative box plots of a quality index at three plants.

Using python to generate box plot

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

BPdata = pd.read_excel('BeesPollen.xlsx')

sns.boxplot(data=BPdata['PercentSlide'])
plt.title('Percent Pollen on Slide')
plt.show()
```



Breakrooms Exercise

**Generate Histogram and
Box Plot (using Python)
For Fruit Prices Data File**

Time Sequence Plots

- A **time series** or **time sequence** is a data set in which the observations are recorded in the order in which they occur
- A **time series plot** is a graph in which the vertical axis denotes the observed value of variable and the horizontal axis denotes the time

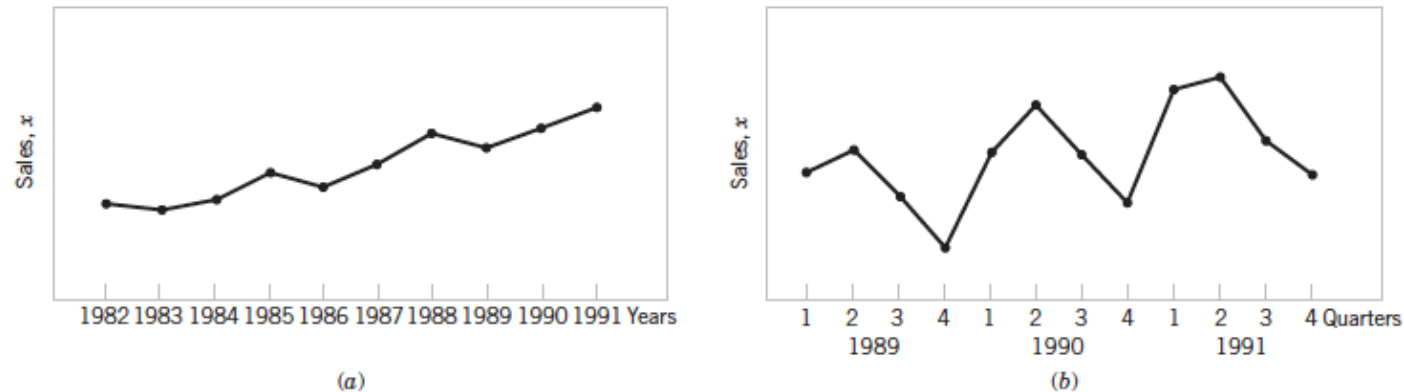


FIGURE 6.16

Company sales by year (a). By quarter (b).

Time Sequence Plots

- Combination of stem – and – leaf plot with a time series plot is a **digidot plot**

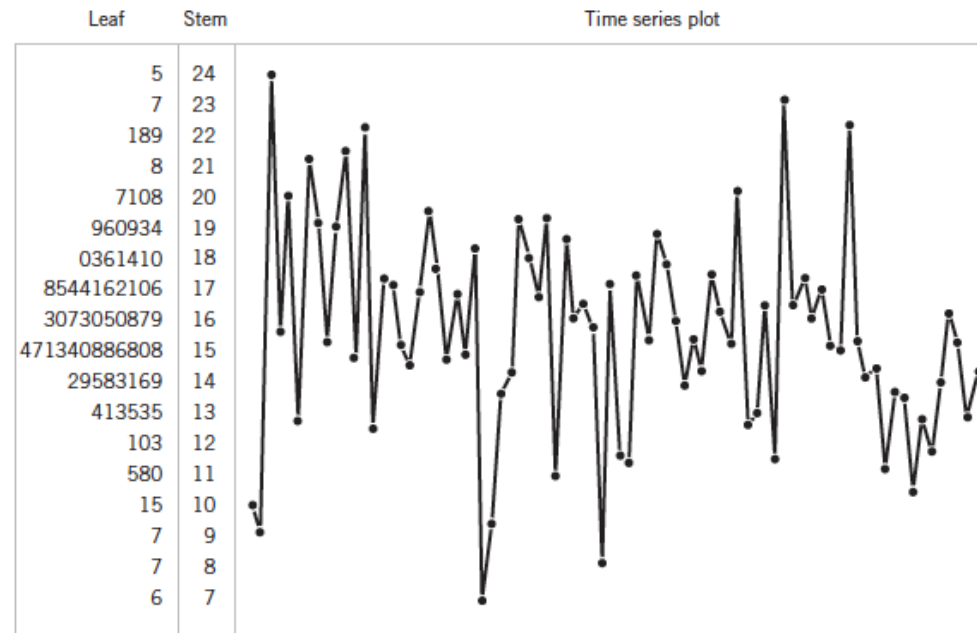


FIGURE 6.17

A digidot plot of the compressive strength data in Table 6.2.

Scatter Diagrams

- **Multivariate**: each observation consists of measurements of several variables
- The **scatter diagram** is a useful way to graphically display the potential relationship between quality and one of the other qualities
- When two or more variables exist, the **matrix of scatter diagrams** may be useful in looking at all of the pairwise relationships between the variables in the sample
- The **sample correlation coefficient** is a quantitative measure of the strength of the linear relationship between two random variables x and y

$$r_{xy} = \frac{\sum_{i=1}^n y_i(x_i - \bar{x})}{\left[\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{1/2}}$$

Sec 6.6 Scatter Diagrams

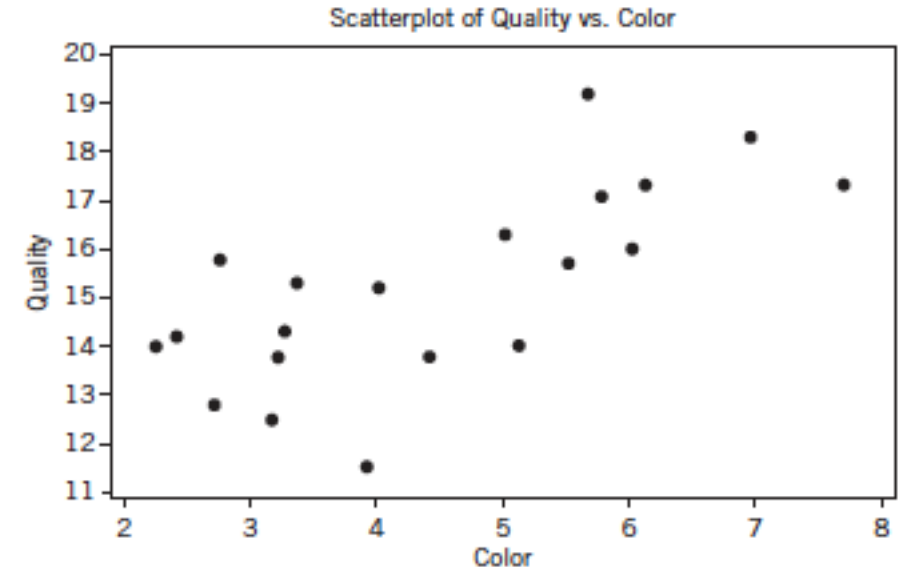


FIGURE 6.19

Scatter diagram of wine quality and color from Table 6.5.

WILEY

Probability Plots

- A **probability plot** is a graphical method for determining whether sample data conform to a hypothesized distribution based on a subjective visual examination of the data
- To construct a probability plot:
 - Rank the data observations in the sample from smallest to largest: $x_{(1)}, x_{(2)}, \dots, x_{(n)}$.
 - The observed value $x_{(j)}$ is plotted against the observed cumulative frequency $(j - 0.5)/n$.
 - The paired numbers are plotted on the probability paper of the proposed distribution.
- If the plotted points deviate a straight line, then the hypothesized distribution adequately describes the data.

Example 6.7 | Battery Life

- The effective service life (X_j in minutes) of batteries used in a laptop are given in the table.
- We hypothesize that battery life is adequately modeled by a normal distribution.
- To test this hypothesis, first arrange the observations in ascending order and calculate their cumulative frequencies and plot them.

j	$x_{(j)}$	$(j - 0.5)/10$	z_j
1	176	0.05	-1.64
2	183	0.15	-1.04
3	185	0.25	-0.67
4	190	0.35	-0.39
5	191	0.45	-0.13
6	192	0.55	0.13
7	201	0.65	0.39
8	205	0.75	0.67
9	214	0.85	1.04
10	220	0.95	1.64

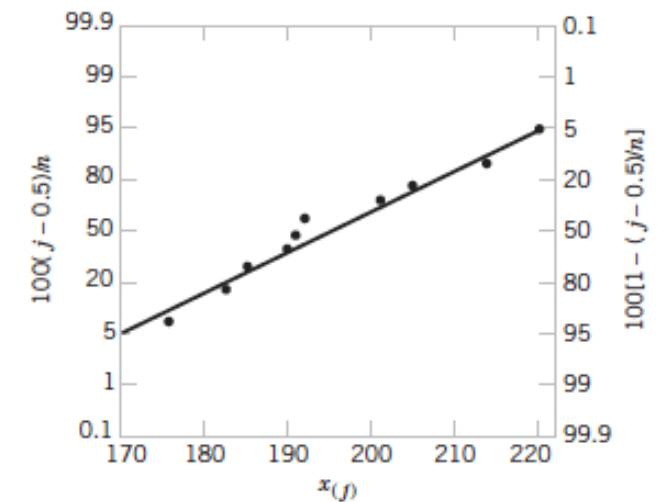


FIGURE 6.22

Normal probability plot for battery life.

Normal Probability Plot

- Can be constructed on ordinary axes by plotting the standardized normal scores z_j against $x(j)$, where the standardized normal scores satisfy

$$\frac{j - 0.5}{n} = P(Z \leq z_j) = \Phi(z_j)$$

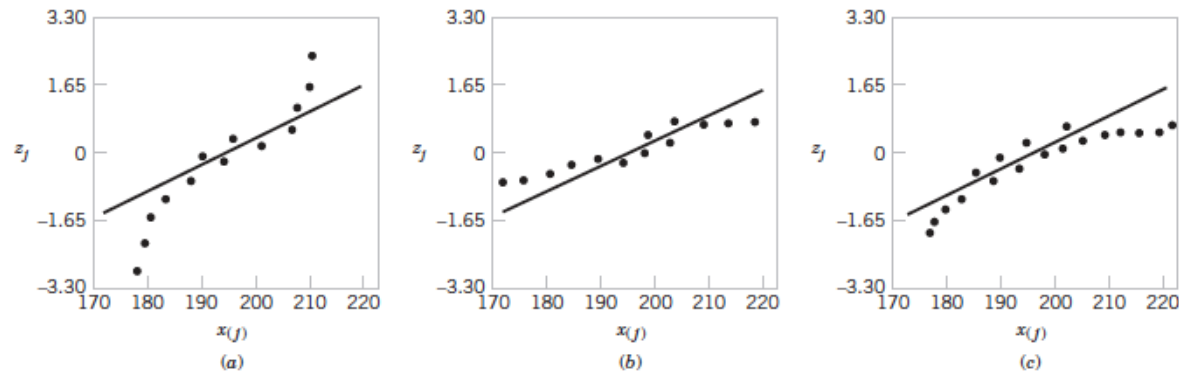


FIGURE 6.24

Normal probability plots indicating a nonnormal distribution. (a) Light-tailed distribution. (b) Heavy-tailed distribution. (c) A distribution with positive (or right) skew.

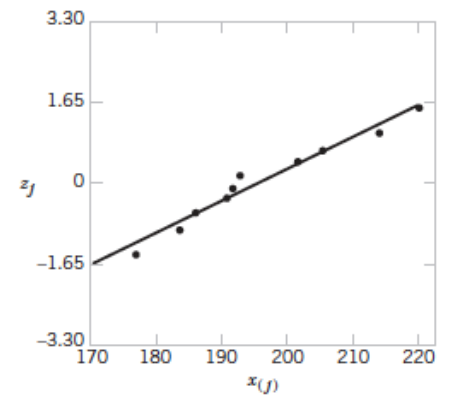


FIGURE 6.23

Normal probability plot obtained from standardized normal scores.

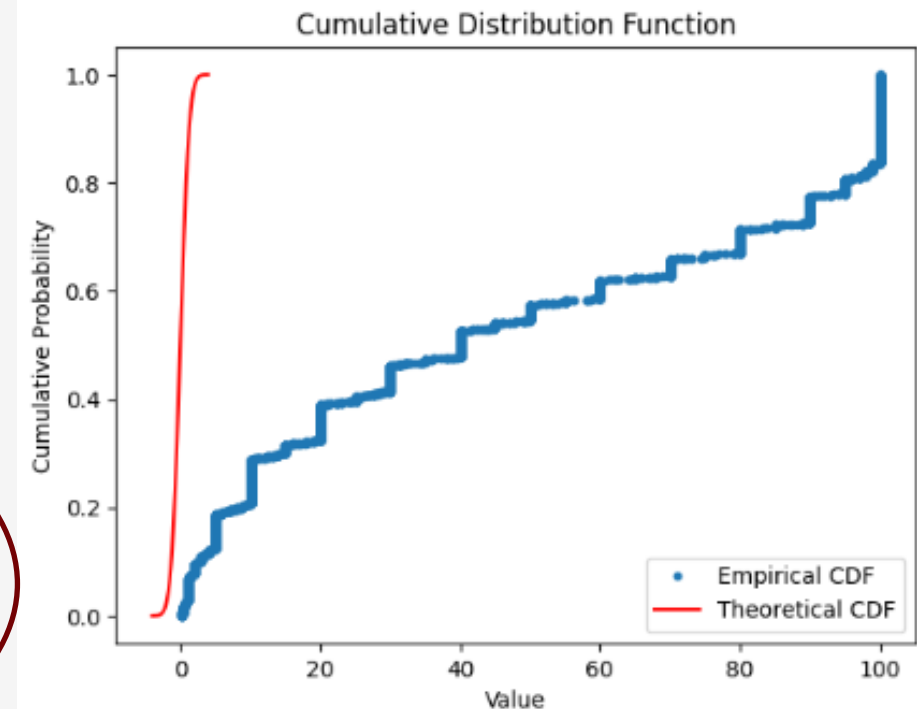
Using Python to Generate Probability Plot

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

BPdata = pd.read_excel('BeesPollen.xlsx')

sorted_data = np.sort(BPdata['PercentSlide'])
cdf = np.arange(1, len(sorted_data) + 1) / len(sorted_data)

plt.plot(sorted_data, cdf, marker='.', linestyle='none', label='Empirical CDF')
x = np.linspace(-4, 4, 1000)
plt.plot(x, norm.cdf(x, loc=8, scale=1), 'r-', label='Theoretical CDF')
plt.title('Cumulative Distribution Function')
plt.xlabel('Value')
plt.ylabel('Cumulative Probability')
plt.legend()
plt.show()
```



Breakrooms Exercise

Can we answer the
following questions (how):

- Average fruit price by product type
- Difference in price for Fresh vs. Canned vs. Juice form of product
- Number of pounds produced by fruit type
- Total sales by fruit type

Semester Project – Deliverable Proposal

- There are plenty of open datasets available for you to use for this project:
 - <https://datasetsearch.research.google.com>
 - <https://www.data.gov/>
 - <https://www.dataquest.io/blog/free-datasets-for-projects/>
- Select at least 3 database files to explore. Perform Descriptive Statistics on all 3 files.
- Based on this, choose which database (1 or more) you will explore for your project.
- Explain why you chose these database
- List initial set of questions you want to answer using this data
- Discuss the initial set of descriptive statistics for this database, and what (if anything) can be inferred from this

Semester Project – Deliverable EDA

- Describe how you used Steps 1 through 8 of EDA process to prepare your data for more detailed analysis.
- Describe what conclusions have you been able to infer, why you inferred them, and what additional questions you are adding to your original list of questions.
- Document should be 5-10 pages, with copy of Python code in an Appendix.

Semester project – Deliverable Podcast

- 10-15 minutes long
- Review you plan/EDA
- Review how you prepped data
- What did you learn from your inference (original questions)
- What did you learn (beyond original questions)
- Advice you would give to somebody going through the same EDA process

Python IDE

- NumPy – numerical computing
- Pandas – clean, filter and manipulate data
- Matplotlib – create basic visualizations
- Seaborn – makes attractive plots
- Scikit-learn – machine learning library with statistical tools
- SciPy – offers advanced statistical functions